C8.3 Barrier responses and domain boundaries

The location of an interconnect in relation to the domain boundaries within a system influences the ability of the interconnect to issue responses to barrier transactions. In general, a system can contain domain boundaries and bi-section boundaries, where:

- a domain boundary is an interface downstream from all master components in the domain
- a bi-section boundary is downstream of a subset of master components in the domain, but not all of them.

Figure C8-1 shows the domain boundaries in an example system.

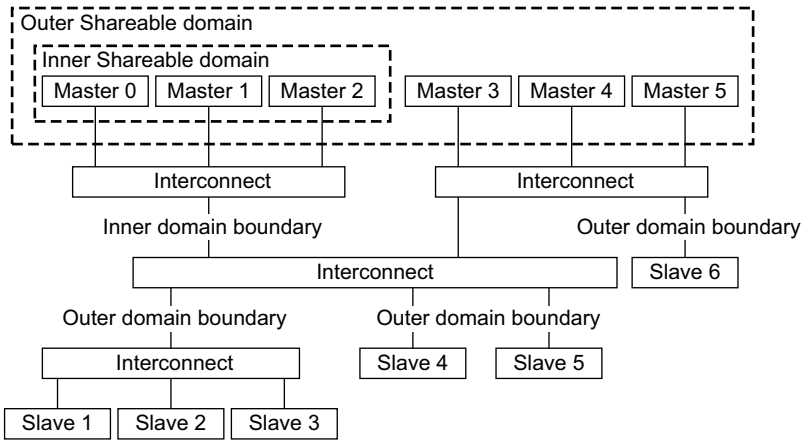


Figure C8-1 Domain boundaries

Note

The interface to slave 6 is an outer domain boundary. Slave 6 cannot be accessed by master 0, master 1, and master 2, and therefore it cannot be considered relative to these master components. It is downstream of all master components in the outer domain that can access it.

Figure C8-2 shows the bi-section boundary locations for the same system.

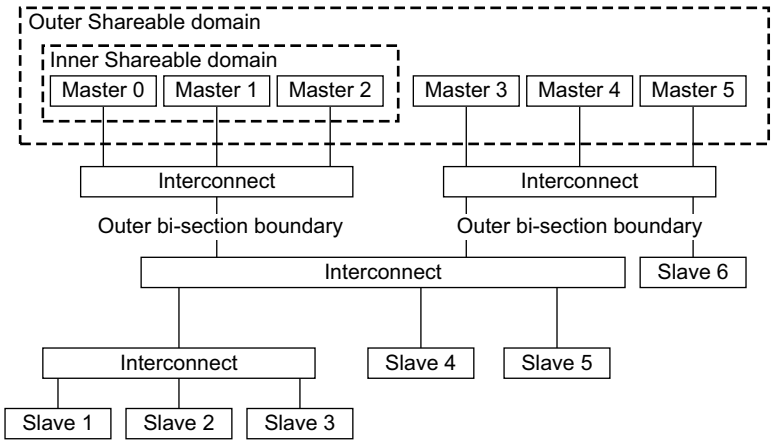


Figure C8-2 Bi-section boundaries

For an interconnect to issue a response to a barrier transaction, certain conditions apply. The main consideration influencing the ability of an interconnect to issue responses to barrier transactions is the location of the interconnect in relation to the domain boundaries within a system.

Copyright © 2003, 2004, 2010, 2011 ARM. All rights reserved.
Non-Confidential

A system can contain the following types of boundary:

Domain boundary

A domain boundary is an interface downstream from all master components in the domain.

For an interface to be a domain boundary, all of the following must apply:

- the set of addresses that pass across the interface is identical for all masters in the domain
- all accesses from any master in the domain to those addresses pass across the interface.

Note

If a master component can access an address using the interface then it must not be possible for another master in the same domain to access the same address without using the interface.

Bi-section boundary

A bi-section boundary is an interface downstream of a subset of master components in the domain, but not all of them.

For an interface to be a bi-section boundary, all of the following must apply:

- the set of addresses that pass across the interface is identical for a subset of master components in the domain
- all accesses from any master component in that subset to those addresses pass across the interface
- no accesses from a master component that is in the domain but is not in the subset passes across the interface
- considering in turn each master component not in the subset, then all addresses that are accessed by both that master and the masters in the subset must be accessed by the masters in the subset across the same interface.

Note

- Informally, an interface is a bi-section boundary if all communication between a subset of masters and the other masters not in the subset pass across the same interface.
- In the definition of a bi-section boundary, the subset of masters is permitted to be all masters in the domain and this makes the bi-section boundary definition the same as the domain boundary definition.

See *Barrier responses and domain boundaries* on page C8-251 for more information.

An interconnect can provide a response to a barrier transaction in certain circumstances. The following rules apply:

- for memory barrier transactions, an interconnect can respond provided it is at the appropriate bi-section boundary or domain boundary, or beyond the domain boundary
- for any synchronization barrier transaction that applies to a Non-shareable, Inner Shareable or Outer Shareable domain, an interconnect can respond provided the interconnect is at or beyond the appropriate domain boundary
- for any synchronization barrier transaction that applies to a System domain, an interconnect can respond provided all transactions before the barrier have reached the endpoint slaves they are destined for.

Copyright © 2003, 2004, 2010, 2011 ARM. All rights reserved.
Non-Confidential

When responding to a barrier transaction, an interconnect must ensure that all transactions that pass across the interface before the barrier are observable to every transaction after the barrier. Some techniques that can be used to achieve this are:

Blocking all transactions and sending barrier

The interconnect blocks all transactions received after the barrier transactions and issues a barrier transaction downstream. The block is removed after a response has been received on both the read data and write response channels for the downstream issued barrier transactions.

Blocking all transactions and waiting for completion

The interconnect blocks all transactions after the barrier transactions and waits for transactions before the barrier to provide a response. The block is removed when all transactions before the barrier have provided a response. To use this technique, it is required that all transactions must have attributes that ensure the response originates from a location that is observable by all masters in the required barrier domain.

Hazard-checking transactions

The interconnect blocks all transactions after the barrier transactions until transactions before the barrier to the same or overlapping addresses have provided a response.

Copyright © 2003, 2004, 2010, 2011 ARM. All rights reserved.
Non-Confidential